

Dilatrend[®]

Carvedilol

1. DESCRIPTION

1.1 Therapeutic / Pharmacologic Class of Drug

Alpha (α) and beta (β) adrenergic receptor blocking agents.

ATC code: C07AG02.

1.2 Type of Dosage Form

Tablets.

The tablet(s) can be divided into equal doses.

1.3 Route of Administration

Oral.

The tablets are to be swallowed with sufficient fluid.

1.4 Sterile / Radioactive Statement

Not applicable.

1.5 Qualitative and Quantitative Composition

Active ingredient: carvedilol.

Tablets

One 6.25 mg tablet contains 6.25 mg carvedilol.

One 25 mg tablet contains 25 mg carvedilol.

The tablets are uncoated, round, and biconvex with a bilateral scorelines, and either yellow (6.25 mg) or white to pale yellowish-beige (25 mg).

2. CLINICAL PARTICULARS

2.1 Therapeutic Indication(s)

Hypertension

Dilatrend is indicated for the management of essential hypertension.

It can be used alone or in combination with other antihypertensive agents (e.g. calcium channel blockers, diuretics).

Treatment of angina pectoris

Dilatrend is indicated for the treatment of angina pectoris.

Chronic heart failure

Unless a contraindication exists, carvedilol is indicated for the treatment of all patients with stable and symptomatic, mild, moderate and severe chronic heart failure of ischemic or non-ischemic etiology in combination with standard therapy (including ACE inhibitors and diuretics with or without digitalis).

2.2 Dosage and Administration

Hypertension

The recommended dose for initiation of therapy is 12.5 mg once a day for the first two days. Thereafter, the recommended dosage is 25 mg once a day. If necessary, the dosage may subsequently be increased at intervals of at least two weeks to the recommended maximum daily dose of 50 mg, given once or twice daily.

Coronary heart disease

The recommended dose for initiation of therapy is 12.5 mg twice a day for the first two days. Thereafter the recommended dosage is 25 mg twice a day. If necessary, the dosage may thereafter be increased at intervals of at least two weeks, up to the recommended maximum daily dose of 100 mg given in divided doses (twice daily).

Chronic heart failure

Dosage must be tailored to suit the individual and closely monitored by a physician during up-titration. For those patients receiving digitalis, diuretics and ACE inhibitors, dosing of these drugs should be stabilized before initiation of carvedilol treatment.

The recommended dose for initiation of therapy is 3.125 mg (half a 6.25 mg tablet) twice daily for two weeks. If

this dose is tolerated, the dose may thereafter be increased, at intervals of not less than two weeks, to 6.25 mg, 12.5 mg and 25 mg twice daily. Doses should be increased to the highest level tolerated by the patient.

The maximum recommended dose is 25 mg twice daily for all patients with severe CHF and for patients with mild to moderate CHF weighing less than 85 kg. In patients with mild or moderate CHF weighing more than 85 kg, the maximum recommended dose is 50 mg twice daily.

Before each dose increase, the patient should be evaluated by the physician for symptoms of vasodilation or worsening heart failure. Transient worsening of heart failure or fluid retention should be treated with increased doses of diuretics. Occasionally, it may be necessary to lower the dose of carvedilol and, in rare cases, temporarily discontinue carvedilol treatment.

If carvedilol treatment is discontinued for more than one week, therapy should be recommenced at a lower dose level (twice daily) and up-titrated in line with the above dosing recommendation. If carvedilol is discontinued for more than two weeks, therapy should be recommenced at 3.125 mg (half a 6.25 mg tablet) in line with the above dosing recommendation.

Symptoms of vasodilation may be managed initially by a reduction in the dose of diuretics. If symptoms persist, the dose of ACE inhibitor (if used) may be reduced, followed by a reduction in the dose of carvedilol if necessary. Under these circumstances, the dose of carvedilol should not be increased until symptoms of worsening heart failure or vasodilation have been stabilized.

Method of administration

The tablets are to be swallowed with sufficient fluid.

Duration of treatment

Treatment with carvedilol is a long-term therapy. As with all β -blockers, treatment should not be stopped abruptly but rather gradually reduced at weekly intervals. This is particularly important in the case of patients with concomitant coronary heart disease.

2.2.1 Special Dosage Instructions

Renal impairment

Available pharmacokinetic data in patients with varying degrees of renal impairment (including renal failure) suggest no changes in carvedilol dosing recommendations are warranted in patients with moderate to severe renal insufficiency.

Hepatic impairment

Carvedilol is contraindicated in patients with clinical manifestations of liver dysfunction (see section 2.3).

Elderly

There is no evidence to support dose adjustment.

2.3 Contraindications

Carvedilol must not be used in patients with:

- Hypersensitivity to carvedilol or any component of the product
- Unstable/decompensated heart failure
- Clinically manifest liver dysfunction

As with other β -blockers, carvedilol must not be used in patients with:

- 2nd and 3rd degree atrioventricular (AV) block (unless a permanent pacemaker is in place)
- severe bradycardia (< 50 bpm)
- sick sinus syndrome (including sino-atrial block)
- severe hypotension (systolic blood pressure < 85 mmHg)
- cardiogenic shock
- history of bronchospasm or asthma

2.4 Warnings and Precautions

2.4.1 General

Chronic congestive heart failure

In congestive heart failure patients, worsening cardiac failure or fluid retention may occur during up-titration of carvedilol. If such symptoms occur, diuretics should be increased and the carvedilol dose should not be further increased until clinical stability resumes. Occasionally, it may be necessary to lower the carvedilol dose or, in rare cases, temporarily discontinue it. Such episodes do not preclude subsequent successful uptitration of carvedilol. Carvedilol should be used with caution in combination with digitalis glycosides, as both drugs slow AV conduction (see section 2.4.3).

Renal function in congestive heart failure

Reversible deterioration of renal function has been observed with carvedilol therapy in chronic heart failure patients with low blood pressure (systolic BP <100 mmHg), ischemic heart disease and diffuse vascular disease, and/or underlying renal insufficiency.

Chronic obstructive pulmonary disease

Carvedilol should be used with caution, in patients with chronic obstructive pulmonary disease (COPD) with a bronchospastic component who are not receiving oral or inhaled medication, and only if the potential benefit outweighs the potential risk.

In patients with a tendency to bronchospasm, respiratory distress can occur as a result of a possible increase in airway resistance. Patients should be closely monitored during initiation and up-titration of carvedilol and the dose of carvedilol should be reduced if any evidence of bronchospasm is observed during treatment.

Diabetes

Care should be taken in the administration of carvedilol to patients with diabetes mellitus, as it may be associated with worsening control of blood glucose. Furthermore, the early signs and symptoms of acute hypoglycaemia may be masked or attenuated (see sections 2.4.3 and 2.5).

In general, β -blockers may increase insulin resistance and mask hypoglycaemic symptoms. However, numerous studies have established that vasodilating β -blockers like carvedilol are associated with more favorable effects on glucose and lipid profiles.

Carvedilol has been shown to exhibit modest insulin-sensitizing properties and can relieve some manifestations of the metabolic syndrome.

Peripheral vascular disease

Carvedilol should be used with caution in patients with peripheral vascular disease (e.g. Raynaud's phenomenon) as β -blockers can precipitate or aggravate symptoms of arterial insufficiency.

Thyrotoxicosis

Carvedilol, like other agents with β -blocking properties, may obscure the symptoms of thyrotoxicosis.

Anesthesia and major surgery

Caution should be exercised in patients undergoing general surgery, because of the synergistic negative inotropic effects of carvedilol and anesthetic drugs (see section 2.4.3).

Bradycardia

Carvedilol may induce bradycardia. If the patient's pulse rate decreases to less than 55 beats per minute, the dosage of carvedilol should be reduced.

Hypersensitivity

Care should be taken in administering carvedilol to patients with a history of serious hypersensitivity reactions, and in patients undergoing desensitisation therapy, as β -blockers may increase both the sensitivity towards allergens and the severity of hypersensitivity reactions.

Risk of Anaphylactic Reaction

While taking beta-blockers, patients with a history of severe anaphylactic reaction to a variety of allergens may be more reactive to repeated challenge, either accidental, diagnostic, or therapeutic. Such patients may be unresponsive to the usual doses of epinephrine used to treat allergic reaction.

Severe cutaneous adverse reactions (SCARs)

Very rare cases of severe cutaneous adverse reactions such as toxic epidermal necrolysis (TEN) and Stevens-Johnson syndrome (SJS) have been reported during treatment with carvedilol [see section 2.6.2 Postmarketing (Undesirable Effects)]. Carvedilol should be permanently discontinued in patients who experience severe cutaneous adverse reactions possibly attributable to carvedilol.

Psoriasis

Patients with a history of psoriasis associated with β -blocker therapy should take carvedilol only after consideration of the risk-benefit ratio.

Interactions with other medicinal products

There are a number of important pharmacokinetic and pharmacodynamic interactions with other drugs (e.g., digoxin, ciclosporin, rifampicin, anaesthetic drugs, anti-arrhythmic drugs), see section 2.4.3 for further details.

Pheochromocytoma

In patients with pheochromocytoma, an α -blocking agent should be initiated prior to the use of any β -blocking agent. Although carvedilol has α - and β -blocking pharmacological activities, there is no experience with its use in this condition. Caution should therefore be taken in the administration of carvedilol to patients suspected of having pheochromocytoma.

Prinzmetal's variant angina

Agents with non-selective β -blocking activity may provoke chest pain in patients with Prinzmetal's variant angina. There is no clinical experience with carvedilol in these patients although the α -blocking activity of carvedilol may prevent such symptoms. Caution should be taken in the administration of carvedilol to patients suspected of having Prinzmetal's variant angina.

Contact lenses

Wearers of contact lenses should bear in mind the possibility of reduced lacrimation.

Withdrawal syndrome

Carvedilol treatment should not be discontinued abruptly, particularly in patients suffering from ischemic heart disease. The withdrawal of carvedilol should be gradual (over a period of two weeks).

2.4.2 Ability to Drive and Use Machines

No studies on the effects on the ability to drive and to use machines have been performed.

Because of individually variable reactions (eg. dizziness, tiredness), the ability to drive, operate machinery, or work without firm support may be impaired. This applies particularly at the start of treatment, after dose increases, on changing products, and in combination with alcohol.

2.4.3 Interactions with Other Medicinal Products and Other forms of Interaction

Pharmacokinetic interactions

Effects of Carvedilol on the pharmacokinetics of other drugs

Carvedilol is a substrate as well as an inhibitor of P-glycoprotein. Therefore the bioavailability of drugs transported by P-glycoprotein may be increased with concomitant administration of carvedilol. In addition, the bioavailability of carvedilol can be modified by inducers or inhibitors of P-glycoprotein.

Digoxin: An increased exposure of digoxin of up to 20% has been shown in some studies in healthy subjects and patients with heart failure. A significantly larger effect has been seen in male patients compared to female patients. Therefore monitoring of digoxin levels is recommended when initiating, adjusting or discontinuing carvedilol (see section 2.4.1). Carvedilol had no effect on digoxin administered intravenously.

Cyclosporin and tacrolimus: Two studies in renal and cardiac transplant patients receiving oral cyclosporin have shown an increase in cyclosporin plasma concentration following the initiation of carvedilol. It appears that carvedilol increases exposure to oral cyclosporin by around 10 to 20%. In an attempt to maintain therapeutic cyclosporin levels, an average 10 to 20% reduction of the cyclosporin dose was necessary. The mechanism for the interaction is not known but inhibition of intestinal P-glycoprotein by carvedilol may be involved. Due to wide interindividual variability of cyclosporin levels, it is recommended that cyclosporin concentrations are monitored closely after initiation of carvedilol therapy and that the dose of cyclosporin be adjusted as appropriate. In case of IV administration of cyclosporin, no interaction with carvedilol is expected. Furthermore, there is evidence that CYP3A4 is involved in the metabolism of carvedilol. As tacrolimus is a substrate of P-glycoprotein and CYP3A4, its pharmacokinetics may also be affected by carvedilol through these interaction mechanisms.

Effects of other drugs and substances on the pharmacokinetics of Carvedilol

Inhibitors as well as inducers of CYP2D6 and CYP2C9 can modify the systemic and/or presystemic metabolism of carvedilol stereoselectively, leading to increased or decreased plasma concentrations of *R*- and *S*-carvedilol (see section 3.2.3). Some examples observed in patients or in healthy subjects are listed below but the list is not exhaustive.

Amiodarone: An in vitro study with human liver microsomes has shown that amiodarone and desethylamiodarone inhibited the oxidation of *R*- and *S*-carvedilol. The trough concentration of *R*- and *S*-carvedilol was significantly increased by 2.2-fold in heart failure patients receiving carvedilol and amiodarone concomitantly as compared to patients receiving carvedilol monotherapy. The effect on *S*-carvedilol was attributed to desethylamiodarone, a metabolite of amiodarone, which is a strong inhibitor of CYP2C9. A monitoring of the β -blockade activity in patients treated with the combination carvedilol and amiodarone is advised.

Rifampicin: In a study in 12 healthy subjects, exposure to carvedilol decreased by around 60% during concomitant

administration with rifampicin and a decrease effect of carvedilol on the systolic blood pressure was observed. The mechanism for the interaction is not known but it may be due to the induction of the intestinal P-glycoprotein by rifampicin. A close monitoring of the β -blockade activity in patients receiving concomitant administration of carvedilol and rifampicin is appropriate.

Fluoxetine and paroxetine: In a randomized, cross-over study in 10 patients with heart failure, co-administration of fluoxetine, a strong inhibitor of CYP2D6, resulted in stereoselective inhibition of carvedilol metabolism with a 77% increase in mean *R*-enantiomer's AUC, and a non-statistically 35% increase of the *S*-enantiomer's AUC as compared to the placebo group. However, no differences in adverse events, blood pressure or heart rate were noted between treatment groups. The effect of single dose paroxetine, a strong CYP2D6 inhibitor, on carvedilol pharmacokinetics was investigated in 12 healthy subjects following single oral administration. Despite significant increase in *R*- and *S*-carvedilol exposure, no clinical effects were observed in these healthy subjects.

Alcohol: Concomitant consumption of alcohol can influence the antihypertensive action of carvedilol and cause different adverse reactions. Alcohol intake was shown to have acute hypotensive effects which possibly augment the blood pressure reduction caused by carvedilol. As *carvedilol* is only sparingly soluble in water but soluble in ethanol, the presence of alcohol could affect the rate and/or extent of intestinal absorption of carvedilol by increasing its solubility. Furthermore, carvedilol was shown to be partly metabolized by CYP2E1, an enzyme known to be both induced and inhibited by alcohol.

Grapefruit juice: Consumption of a single dose of 300 mL grapefruit juice was shown to result in a 1.2-fold increase of the AUC of carvedilol in comparison to water. While the clinical relevance of this observation is unclear, it is advisable that patients should avoid concomitant intake of grapefruit juice at least until a stable dose-response relationship is established.

Pharmacodynamic interactions

Insulin or oral hypoglycaemics: Agents with β -blocking properties may enhance the blood-sugar-reducing effect of insulin and oral hypoglycaemics. The signs of hypoglycaemia may be masked or attenuated (especially tachycardia). In patients taking insulin or oral hypoglycaemics, regular monitoring of blood glucose is therefore recommended (see section 2.4.1 General).

Catecholamine-depleting agents: Patients taking both agents with β -blocking properties and a drug that can deplete catecholamines (e.g. reserpine and monoamine oxidase inhibitors) should be observed closely for signs of hypotension and/or severe bradycardia.

Digoxin: The combined use of β -blockers and digoxin may result in additive prolongation of atrioventricular (AV) conduction time (see section 2.4.1 General).

Non-dihydropyridines calcium channel blockers, amiodarone or other antiarrhythmics: In combination with carvedilol can increase the risk of AV conduction disturbances (see section 2.4.1 Warnings and Precautions). Isolated cases of conduction disturbance (rarely with haemodynamic compromise) have been observed when carvedilol is co-administered with diltiazem. As with other agents with β -blocking properties, if carvedilol is to be administered orally with non-dihydropyridines calcium channel blockers of the verapamil or diltiazem type, amiodarone or other antiarrhythmics it is recommended that ECG and blood pressure be monitored.

Clonidine: Concomitant administration of clonidine with agents with β -blocking properties may potentiate blood pressure- and heart-rate-lowering effects. When concomitant treatment with agents with β -blocking properties and clonidine is to be terminated, the β -blocking agent should be discontinued first. Clonidine therapy can then be discontinued several days later by gradually decreasing the dosage.

Antihypertensives: As with other agents with β -blocking activity, carvedilol may potentiate the effect of other concomitantly administered drugs that are anti-hypertensive in action (e.g. α_1 -receptor antagonists) or have hypotension as part of their adverse effect profile.

Anesthetic agents: Careful monitoring of vital signs is recommended during anaesthesia due to the synergistic negative inotropic and hypotensive effects of carvedilol and anaesthetic drugs.

Nonsteroidal anti-inflammatory drugs (NSAIDs): The concurrent use of non-steroidal anti-inflammatory drugs (NSAIDs) and β -adrenergic blockers may result in an increase in blood pressure and impairment of blood pressure control.

β-agonist bronchodilators: Non-cardioselective *β*-blockers oppose the bronchodilator effects of *β*-agonist bronchodilators. Careful monitoring of patients is recommended.

2.5 Use in Special Populations

2.5.1 Pregnancy

Studies in animals have shown reproductive toxicity (see section 3.3). The potential risk for humans is unknown. *β*-blockers reduce placental perfusion, which may result in intrauterine fetal death, and immature and premature deliveries. In addition, adverse effects (especially hypoglycaemia and bradycardia) may occur in the fetus and neonate. There may be an increased risk of cardiac and pulmonary complications in the neonate in the postnatal period. There is no evidence from animal studies that carvedilol has any teratogenic effects.

There is no adequate clinical experience with carvedilol in pregnant women. Carvedilol should not be used during pregnancy unless the potential benefit outweighs the potential risk.

2.5.2 Nursing Mothers

Animal studies demonstrated that carvedilol and/or its metabolites are excreted in rat breast milk. The excretion of carvedilol in human milk has not been established. However, most *β*-blockers, in particular lipophilic compounds, will pass into human breast milk although to a variable extent. Breast feeding is therefore not recommended following administration of carvedilol.

2.5.3 Geriatric Use

A study in elderly hypertensive patients showed that there was no difference in the adverse event profile as compared to younger patients. Another study, which included elderly patients with coronary heart disease, showed no difference in the adverse events reported vs those reported by younger patients. Therefore, no dose adjustment of the starting dose is required in the geriatric population.

2.5.4 Renal Impairment

The autoregulatory renal blood supply is preserved and the glomerular filtration is unchanged during chronic treatment with carvedilol. In patients with moderate to severe renal insufficiency, no changes in carvedilol dosage recommendations are warranted.

2.5.5 Hepatic Impairment

Carvedilol is contraindicated in patients with clinically manifest liver dysfunction (see section 2.3). A pharmacokinetic study in cirrhotic patients has shown that exposure (AUC) to carvedilol was increased by 6.8-folds in patients with liver impairment as compared to healthy subjects.

2.6 Undesirable Effects

2.6.1 Clinical Trials

Adverse Drug Reactions (ADRs) are listed according to MedDRA system organ class and CIOMS Frequency category:

Very common	≥ 1/10
Common	≥ 1/100 and < 1/10
Uncommon	≥ 1/1,000 and < 1/100
Rare	≥ 1/10,000 and < 1/1,000
Very rare	< 1/10,000

The table 1 below summarizes undesirable effects that have been reported in association with the use of carvedilol in pivotal clinical trials with following indications: chronic heart failure, left ventricular dysfunction following acute myocardial infarction, hypertension and the long term management of coronary heart disease

Table 1 Adverse drug reactions in clinical trials

System Organ Class	Adverse reaction	Frequency
Infections and infestations	Pneumonia	Common
	Bronchitis	Common
	Upper respiratory tract infection	Common
	Urinary tract infection	Common
Blood and lymphatic system disorders	Anaemia	Common
	Thrombocytopenia	Rare
	Leukopenia	Very rare
Immune system disorders	Hypersensitivity (allergic reactions)	Very rare
Metabolism and nutrition disorders	Hypercholesterolaemia	Common
	Impaired blood glucose control (hyperglycaemia, hypoglycaemia) in patients with pre-existing diabetes	Common
Psychiatric disorders	Depression, depressed mood	Common
	Sleep disorder	Uncommon
Nervous system disorders	Dizziness	Very common
	Headache	Very Common
	Syncope, presyncope	Common
	Paraesthesia	Uncommon
Eye disorders	Visual impairment	Common
	Lacrimation decreased (dry eye)	Common
	Eye irritation	Common
Cardiac disorders	Cardiac failure	Very common
	Bradycardia	Common
	Hypervolaemia (fluid overload)	Common
	Atrioventricular block	Uncommon
	Angina pectoris	Uncommon
Vascular disorders	Hypotension	Very common
	Orthostatic hypotension	Common
	Disturbances of peripheral circulation (cold extremities, peripheral vascular disease, exacerbation of intermittent claudication and Raynaud's phenomenon)	Common
	Hypertension	Common
Respiratory, thoracic and mediastinal disorders	Dyspnoea	Common
	Pulmonary oedema	Common
	Asthma in predisposed patients	Common
	Nasal congestion	Rare
Gastrointestinal disorders	Nausea	Common
	Diarrhoea	Common
	Vomiting	Common
	Dyspepsia	Common
	Abdominal pain	Common
	Constipation	Uncommon
	Dry mouth	Rare
Skin and subcutaneous disorders	Skin reactions (e.g. allergic exanthema, dermatitis, urticaria, pruritus, psoriatic plaque and lichen planus like eruptions)	Uncommon
Musculoskeletal and connective tissue disorders	Pain in extremities	Common
Renal and urinary disorders	Renal failure and renal impairment in patients with diffuse vascular disease and/or underlying renal insufficiency	Common
	Micturition disorders	Rare

System Organ Class	Adverse reaction	Frequency
Reproductive system and breast disorders	Erectile dysfunction	Uncommon
General disorders and administration site conditions	Asthenia (fatigue)	Very common
	Oedema	Common
	Pain	Common
Investigations	Weight increase	Common
	Alanine aminotransferase (ALT), aspartate aminotransferase (AST) and gamma-glutamyltransferase (GGT) increased	Very rare

Description of selected adverse reactions

The frequency of adverse reactions is not dose-dependent, with the exception of dizziness, abnormal vision and bradycardia. Dizziness, syncope, headache and asthenia are usually mild and are more likely to occur at the beginning of treatment.

In patients with congestive heart failure, worsening cardiac failure and fluid retention may occur during up-titration of carvedilol dose (see section 2.4).

Cardiac failure was a very commonly reported adverse event in both placebo (14.5%) and carvedilol-treated (15.4%) patients, in patients with left ventricular dysfunction following acute myocardial infarction.

Reversible deterioration of renal function has been observed with carvedilol therapy in chronic heart failure patients with low blood pressure, ischaemic heart disease and diffuse vascular disease and/or underlying renal insufficiency (see section 2.4).

2.6.2 Post Marketing

The following adverse events have been identified during post-marketing use of carvedilol. Because these events are reported from a population of uncertain size, it is not always possible to reliably estimate their frequency and/or establish a causal relationship to drug exposure.

Cardiac disorders

Sinus arrest in predisposed patients (e.g. elderly patients or patients with pre-existing bradycardia, sinus node dysfunction or atrioventricular block).

Metabolism and nutrition disorders

Due to the β -blocking properties, it is also possible for latent diabetes mellitus to become manifest, manifest diabetes to be aggravated, and blood glucose counter-regulation to be inhibited.

Psychiatric disorders

Hallucination.

Renal and urinary disorders

Isolated cases of urinary incontinence in women, which resolved upon discontinuation of the medication, have been reported.

Skin and subcutaneous tissue disorders

Alopecia

Severe cutaneous adverse reactions (Toxic epidermal necrolysis, Stevens-Johnson syndrome (see section 2.4 Warnings and Precautions).

Hyperhidrosis.

2.7 Overdose

Symptoms and signs of overdose

In the event of overdosage, there may be severe hypotension, bradycardia, heart failure, cardiogenic shock, sinus arrest and cardiac arrest. There may also be respiratory problems, bronchospasm, vomiting, disturbed consciousness and generalized seizures.

Treatment of overdose

The patients should be monitored for the above mentioned signs and symptoms and managed according to the best judgment of the treating physicians and according to standard practice for patients with β -blocker overdose (e.g. atropine, transvenous pacing, glucagon, phosphodiesterase inhibitor such as amrinone or milrinone, β -sympathomimetics).

Important note

In cases of severe intoxication with shock, supportive treatment must be continued for a sufficiently long period, as a prolongation of elimination half-life and redistribution of carvedilol from deeper compartments are to be expected. The duration of the supportive/antidote therapy depends on the severity of the overdose. The supportive treatment should therefore be continued until the patient's condition has stabilized.

3. PHARMACOLOGICAL PROPERTIES AND EFFECTS

3.1 Pharmacodynamic Properties

Pharmacotherapeutic group: Alpha (α) and Beta (β) adrenergic receptor blocking agents, ATC code: C07AG02.

3.1.1 Mechanism of Action

Carvedilol, a racemic mixture of two enantiomers (*R*- and *S*-carvedilol), is a multiple action α - and β -adrenergic receptor blocker. The β -adrenergic receptor blockade is associated with the *S*-enantiomer and non-selective for β_1 - and β_2 -adrenoceptors, while both enantiomers have the same blocking properties specific for α_1 -adrenergic receptors. At higher concentrations, carvedilol also has a weak to moderate calcium-channel blocking activity. It has no intrinsic sympathomimetic activity and (like propranolol) it has membrane-stabilising properties.

3.1.2 Pharmacodynamic effects

In addition to its cardiovascular effects resulting from β -adrenergic receptor blockade as described in more detail in the following section, carvedilol reduces peripheral vascular resistance via its selective blockade of α_1 -adrenoceptors. Furthermore, its calcium-channel blocking activity may increase blood flow in specific vascular beds such as the cutaneous circulation. Through its β -blocking action, carvedilol suppresses the renin-angiotensin-aldosterone system, thereby reducing the release of renin and making fluid retention rare. It attenuates the increase in blood pressure induced by phenylephrine, an α_1 -adrenoceptor agonist, but not that induced by angiotensin II.

Carvedilol has also been shown to have organ-protective effects likely resulting at least in part from additional properties beyond its adrenergic receptor blockade action. It has potent antioxidant properties associated with both enantiomers, and is a scavenger of reactive oxygen radicals. A decrease in oxidative stress has been shown in clinical studies by measuring various markers during chronic treatment of patients with carvedilol. Furthermore, it has antiproliferative effects on human vascular smooth muscle cells.

Carvedilol has no adverse effect on the lipid profile. A normal ratio of high-density lipoproteins to low density lipoproteins (HDL/LDL) is maintained.

3.1.3 Clinical / Efficacy Studies

Clinical studies showed the following results for carvedilol:

Hypertension

Carvedilol lowers blood pressure in hypertensive patients by a combination of β -blockade and α_1 mediated vasodilation. Some of the limitations of traditional β -blockers do not appear to be shared by some of the vasodilating β -blockers, such as carvedilol. A reduction in blood pressure is not associated with a concomitant increase in total peripheral resistance, as observed with pure β -blocking agents. Heart rate is slightly decreased. Renal blood flow and renal function are maintained in hypertensive patients. Carvedilol has been shown to maintain stroke volume and reduce total peripheral resistance. Blood supply to distinct organs and vascular beds including kidneys, skeletal muscles, forearms, legs, skin, brain or the carotid artery is not compromised by carvedilol. There is a reduced incidence of cold extremities and early fatigue during physical activity. The long-term effect of carvedilol on hypertension is documented in several double-blind controlled studies.

Hypertensive patients with renal impairment

Several open studies have shown that carvedilol is an effective agent in patients with renal hypertension. The same is true in patients with chronic renal failure or those on haemodialysis or after renal transplantation. Carvedilol causes a gradual reduction in blood pressure both on dialysis and non-dialysis days, and the blood pressure-lowering effects are comparable with those seen in patients with normal renal function.

On the basis of results obtained in comparative trials on haemodialysed patients, it was concluded that carvedilol was more effective than calcium channel blockers and was better tolerated.

Coronary Heart Disease

In patients with coronary heart disease, carvedilol has demonstrated anti-ischemic (improved total exercise time, time to 1mm ST segment depression and time to angina) and anti-anginal properties that were maintained during long-term treatment. Acute hemodynamic studies have demonstrated that carvedilol significantly decreases myocardial oxygen demand and sympathetic overactivity. It also decreases the myocardial preload (pulmonary artery pressure and pulmonary capillary wedge pressure) and afterload (total peripheral resistance).

Chronic Heart Failure

Carvedilol significantly reduces mortality and hospitalisations and improves symptoms and left ventricular function in patients with ischemic or non-ischemic chronic heart failure. The effect of carvedilol is dose dependent.

Chronic Heart Failure patients with renal impairment

Carvedilol reduces morbidity and mortality in dialysis patients with dilated cardiomyopathy, as well as all-cause mortality, cardiovascular mortality and heart failure mortality or first hospitalization in heart failure patients with mild to moderate non-dialysis-dependent chronic kidney disease. A meta-analysis of placebo-controlled clinical trials including a large number of patients (> 4,000) with mild to moderate chronic kidney disease supports carvedilol treatment of patients with left ventricular dysfunction with or without symptomatic heart failure to reduce rates of all cause of mortality as well as heart failure related events.

Paediatric population

Available studies regarding safety and efficacy in children and adolescents are limited in number and size, and were focused on treatment of paediatric heart failure which however differs from the disease in adults regarding characteristics and aetiology. While a number of preliminary and observational investigations concerning this condition including studies on heart failure secondary to muscular dystrophy reported possible beneficial effects of carvedilol, evidence regarding efficacy from randomized controlled trials is conflicting and inconclusive.

The safety data from these studies indicates that adverse events were generally comparable between carvedilol-treated and control groups. However, because of the small number of participants compared to studies in adults and a general lack of an optimal dosing scheme for children and adolescents, the available data is not sufficient to establish a paediatric safety profile for carvedilol. Use of carvedilol in paediatric patients is therefore a safety concern and not recommended as substantial information regarding benefits and risks is missing.

3.2 Pharmacokinetic Properties

3.2.1 Absorption

Following oral administration of a 25 mg capsule to healthy subjects, carvedilol is rapidly absorbed with a peak plasma concentration C_{max} of 21 µg/L reached after approximately 1.5 hour (t_{max}). The C_{max} values are linearly related to the dose. Following oral administration, carvedilol undergoes extensive first pass metabolism that results in an absolute bioavailability of about 25% in healthy male subjects. Carvedilol is a racemate and the *S*- enantiomer appears to be metabolized more rapidly than the *R*-enantiomer, showing an absolute oral bioavailability of 15% compared to 31% for the *R*-enantiomer. The maximal plasma concentration of *R*-carvedilol is approximately twofold higher than that of *S*-carvedilol.

In vitro studies have shown that carvedilol is a substrate of the efflux transporter P-glycoprotein. The role of P-glycoprotein in the disposition of carvedilol was also confirmed in vivo in healthy subjects.

3.2.2 Distribution

Carvedilol is a highly lipophilic compound, showing a plasma protein binding of around 95%. The distribution volume ranges between 1.5 and approximately 2 L/kg.

3.2.3 Metabolism

In humans, carvedilol is extensively metabolized in the liver via oxidation and conjugation into a variety of metabolites that are eliminated mainly in the bile. Enterohepatic circulation of the parent substance has been shown in animals.

Demethylation and hydroxylation at the phenol ring produce 3 metabolites with β-adrenergic receptor blocking activity. Based on preclinical studies, the 4'-hydroxyphenol metabolite is approximately 13 times more potent than carvedilol for β-blockade. Compared to carvedilol, the three active metabolites exhibit weak vasodilating activity. In humans, the concentrations of the three active metabolites are about 10 times lower than that of the parent substance. Two of the hydroxy-carbazole metabolites of carvedilol are extremely potent antioxidants, demonstrating a 30 to 80 fold greater potency than carvedilol.

Pharmacokinetic studies in human have shown that the oxidative metabolism of carvedilol is stereoselective. The results of an in vitro study suggested that different cytochrome P450 isoenzymes may be involved in the oxidation and hydroxylation processes including CYP2D6, CYP3A4, CYP2E1, CYP2C9, as well as CYP1A2.

Studies in healthy volunteers and in patients have shown that the *R*-enantiomer is predominantly metabolized by CYP2D6. The *S*-enantiomer is mainly metabolized by CYP2D6 and CYP2C9.

Genetic polymorphism

The results of clinical pharmacokinetic studies in human subjects have shown that CYP2D6 plays a major role in the metabolism of *R*- and of *S*-carvedilol. As a consequence plasma concentrations of *R*- and *S*-carvedilol are increased in CYP2D6 slow metabolisers. The importance of CYP2D6 genotype in the pharmacokinetics of *R*- and *S*-carvedilol was confirmed in population pharmacokinetics studies, whereas other studies did not confirm this observation. It was concluded that CYP2D6 genetic polymorphism may be of limited clinical significance. This assessment is also supported by the observations that differences in pharmacokinetic characteristics due to

CYP2D6 polymorphism did not have significant effects on the pharmacodynamic response in healthy volunteers, and that there was no association between CYP2D6 genotype or phenotype and carvedilol dose or rate of adverse drug reactions in heart failure patients.

3.2.4 Elimination

Following a single oral administration of 50 mg carvedilol, around 60% are secreted into the bile and eliminated with the faeces in the form of metabolites within 11 days. Following a single oral dose, only about 16% are excreted into the urine in form of carvedilol or its metabolites. The urinary excretion of unaltered drug represents less than 2%. After intravenous infusion of 12.5 mg to healthy volunteers, the plasma clearance of carvedilol reaches around 600 mL/min and the elimination half-life around 2.5 hours. The elimination half-life of a 50 mg capsule observed in the same individuals was 6.5 hours corresponding indeed to the absorption half-life from the capsule. Following oral administration, the total body clearance of the *S*-carvedilol is approximately two times larger than that of the *R*-carvedilol.

3.2.5 Pharmacokinetic/Pharmacodynamic Relationship(s)

Adrenergic receptor binding of carvedilol as examined by radioreceptor assay and its enantiomer kinetics were found to be proportional with dose, with an overall smooth linear relationship between the treatment response measured by ergometric heart rate increase and log-transformed dose, log-transformed concentrations of the *S*-enantiomer, and β_1 -adrenergic receptor occupancies. Similar to other β -blockers, there is a temporal disconnection between drug exposure and pharmacodynamic action as maximal effect regarding heart rate and blood pressure is achieved later than maximal plasma concentration. Metabolic differences caused by distinct CYP2D6 genotypes were found to result in significant pharmacokinetic variations but to have no effect on heart rate, blood pressure or adverse effects, likely because of compensation by active metabolites and an overall flat concentration-effect relationship. Altogether, dose-effect and kinetic-dynamic relationships of carvedilol are determined by complex interactions between enantiospecific kinetics and dynamics, protein binding, and involvement of active metabolites as outlined before in this section.

3.2.6 Pharmacokinetics in Special Populations

Renal impairment

In patients with hypertension and renal insufficiency, the area under plasma level-time curve, elimination half-life and maximum plasma concentration does not change significantly. Renal excretion of the unchanged drug decreases in the patients with renal insufficiency; however, changes in pharmacokinetic parameters are modest. Carvedilol is not eliminated during dialysis because it does not cross the dialysis membrane, probably due to its high plasma protein binding.

Hepatic impairment

See section 2.3 and 2.5.5.

Hepatic failure

In a study in 24 Japanese patients with heart failure, the clearance of *R*- and *S*-carvedilol was significantly lower than previously estimated in healthy volunteers. These results suggested that the pharmacokinetics of *R*- and *S*-carvedilol is significantly altered by heart failure.

Elderly

Age has no statistically significant effect on the pharmacokinetics of carvedilol in hypertensive patients.

Children

The weight-adjusted clearance in children and adolescents was shown to be significantly larger than in adults.

3.3 Preclinical Safety Data

3.3.1 Carcinogenicity

In carcinogenicity studies conducted in rats and mice, employing dosages up to 75 mg/kg/day and 200 mg/kg/day respectively (38 to 100 times the maximum recommended human dose [MRHD]), carvedilol had no carcinogenic effect.

3.3.2 Mutagenicity

Carvedilol was not mutagenic in *in vitro* or *in vivo* mammalian tests and non-mammalian tests.

3.3.3 Impairment of Fertility

Administration of carvedilol to adult female rats at toxic doses (≥ 200 mg/kg, ≥ 100 times MRHD) resulted in impairment of fertility (poor mating, fewer corpora lutea and fewer implants).

3.3.4 Teratogenicity

There is no evidence from animal studies that carvedilol has any teratogenic effects. Doses > 60 mg/kg (> 30 times MRHD) caused delays in physical growth/development of offspring. There was embryotoxicity (increased post-implantation deaths) but no malformations in rabbits and rats at doses of 75 mg/kg and 200 mg/kg, respectively (38 to 100 times MRHD).

4. PHARMACEUTICAL PARTICULARS

4.1 Storage

This medicine should not be used after the expiry date (EXP) shown on the pack.

Do not store above 30 °C. Store in the original container (sensitive to light).

4.2 Special Instructions for Use, Handling and Disposal

The release of pharmaceuticals in the environment should be minimized. Medicines should not be disposed of via wastewater and disposal through household waste should be avoided. Use established “collection systems”, if available in your location.

4.3 Packs

Tablets 6.25 mg.....30's, 100's

Tablets 25 mg.....30's, 100's

Medicine: keep out of reach of children

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